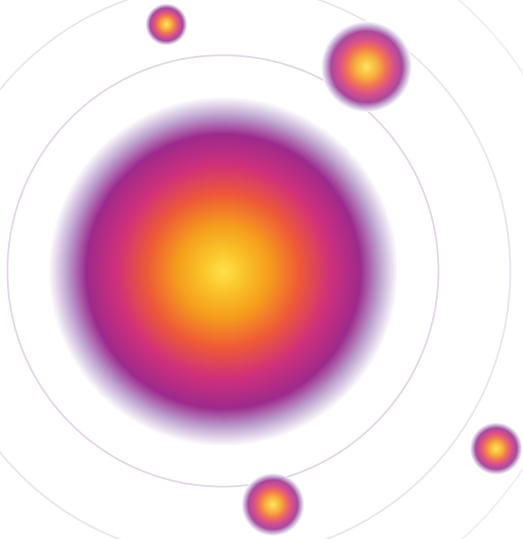
R&D Imaging Seminar · 3 November 2026

Handling the inversion of a severely ill-conditioned operator: the case of fDOT

Fluorescence diffuse optical tomography (fDOT) for preclinical small-animal imaging



Clément Lavaud

Anabela Da Silva · Caroline Chaux · Martin Rodriguez-Vega · Julien Wojak · Rémi André · Pascal Nguyen

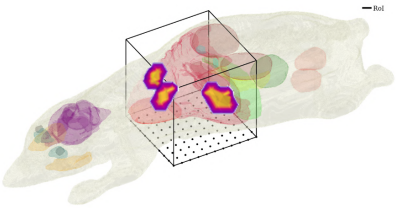
Aix-Marseille Université · I2M · Institut Fresnel
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How can we stably invert a severely ill-conditioned fDOT operator?



Context: why fDOT for small-animal imaging?

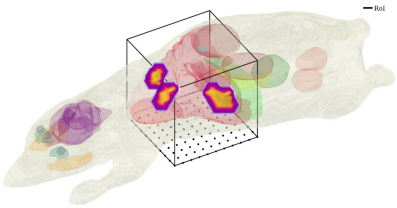
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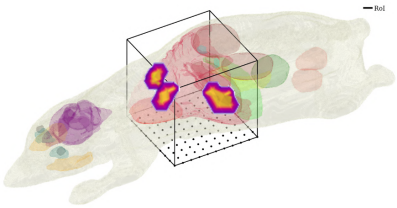
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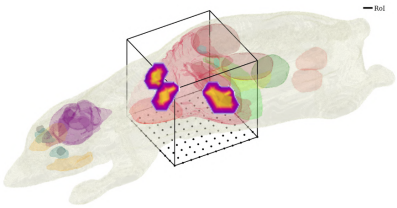
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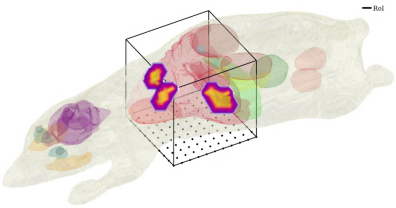
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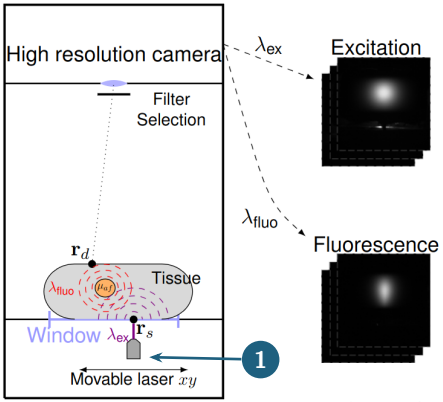
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- **From 2D to 3D. fDOT** reconstructs the fluorophore location and concentration in 3D.
- **The challenge.** Strong light scattering makes reconstruction ill-posed.



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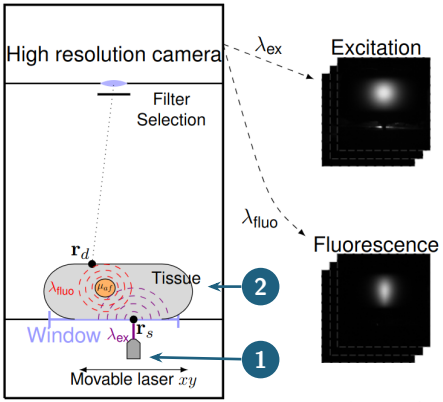
Acquisition setup of fDOT



Physical principle

1. **Excitation.** A NIR-I **movable laser** ($\lambda_{\text{ex}} \simeq 808\,\text{nm}$) illuminates the animal from below, at several source positions $\mathbf{r}_s$.

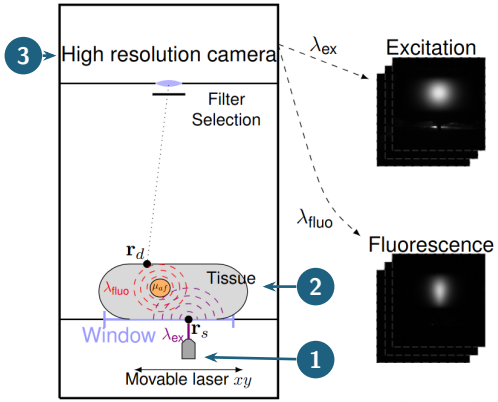
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3. **Detection.** A **high-resolution camera** above the animal records two light-intensity measurements, at λ_{ex} and λ_{fluo}, for each source–detector pair $(\mathbf{r}_s, \mathbf{r}_d)$.

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Forward model: from the RTE to the diffusion equation

Light propagation in a scattering medium is described by the **radiative transfer equation** (RTE) for the radiance $L(\mathbf{r}, \mathbf{u})$:

Radiative transfer equation (RTE) [Ishimaru 1978]

$$\mathbf{u} \cdot \nabla L(\mathbf{r}, \mathbf{u}) + (\mu_a(\mathbf{r}) + \mu_s(\mathbf{r}))L(\mathbf{r}, \mathbf{u}) = \mu_s(\mathbf{r}) \int_{4\pi} \Theta(\mathbf{u} \cdot \mathbf{u}')L(\mathbf{r}, \mathbf{u}')\, d^2\mathbf{u}' + S(\mathbf{r}, \mathbf{u})$$

$\mathbf{u}$: propagation direction; μ_a, μ_s: absorption and scattering coefficients; Θ: scattering phase function; S: source term.

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In highly scattering tissue at the wavelengths of interest, $\mu_s' \gg \mu_a$ (μ_s': reduced scattering coefficient). The RTE then reduces to the **diffusion equation** for the fluence rate $\Phi(\mathbf{r})$:

Diffusion equation [Arridge 1999]

$$-\nabla \cdot \left[D(\mathbf{r})\, \nabla \Phi(\mathbf{r})\right] + \mu_a(\mathbf{r})\, \Phi(\mathbf{r}) = S(\mathbf{r}), \qquad D = \frac{1}{3(\mu_a + \mu_s')}$$

Coupled diffusion equations

In fDOT, the diffusion equation is solved at **two** wavelengths, coupled through the fluorophore:

Coupled excitation / emission system

$$\begin{cases} -\nabla\cdot\left(D_{\mathrm{ex}}\nabla\Phi_{\mathrm{ex}}\right)+\mu_{a,\mathrm{ex}}\Phi_{\mathrm{ex}}=S, \\ -\nabla\cdot\left(D_{\mathrm{em}}\nabla\Phi_{\mathrm{em}}\right)+\mu_{a,\mathrm{em}}\Phi_{\mathrm{em}}=\eta\,\mu_{af}(\mathbf{r})\Phi_{\mathrm{ex}}. \end{cases}$$

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$\Phi_{\mathrm{ex}}, \Phi_{\mathrm{em}}$: excitation/emission fluence rates.

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The emission equation is **linear** in μ_{af} once Φ_{ex} is fixed (the fluorophore is assumed not to perturb the excitation field, i.e. Born approximation). Discretization then yields a linear system.

Linear inverse problem

After discretization, the **forward model** links the fluorophore distribution to the measured signal:

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Building A and m (normalized Born) [Ntziachristos & Weissleder 2001]

$$m_j = \frac{U^{\mathrm{fluo}}(\mathbf{r}_s^j, \mathbf{r}_d^j)}{U^{\mathrm{ex}}(\mathbf{r}_s^j, \mathbf{r}_d^j)}$$

$$A_{j,i} = \frac{G^{\mathrm{ex}}(\mathbf{r}_s^j, \mathbf{r}^i)\, G^{\mathrm{fluo}}(\mathbf{r}^i, \mathbf{r}_d^j)}{G^{\mathrm{ex}}(\mathbf{r}_s^j, \mathbf{r}_d^j)}$$

U: intensities measured at $\mathbf{r}_d$ (the ratio removes unknown source/detector gains); G: Green's functions.

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A severely ill-posed problem

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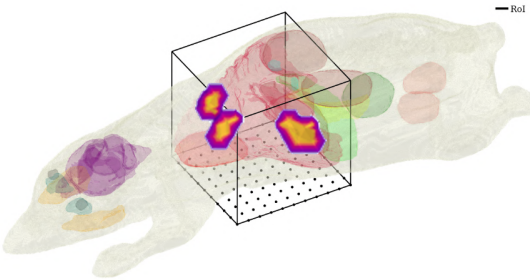
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Data [Rodriguez-Vega et al. 2026]

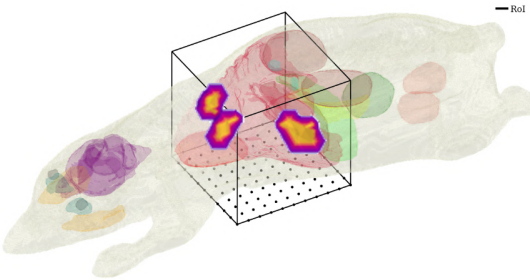
- Realistic simulated small-animal database (NIR fDOT)
- Various geometries for fluorophore inclusions
- Known ground truth $\mathbf{c}^*$ for evaluation
- Dimensions of $(\mathbf{m}, \mathbf{A})$ vary with the geometry

Simulated phantom and reference inclusion

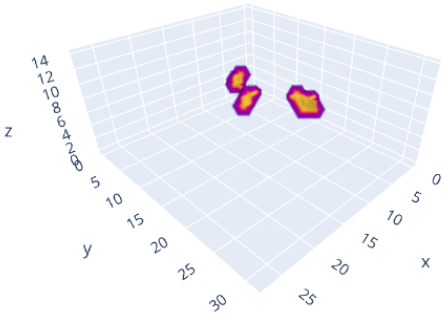


Simulated mouse with RoI and fluorescent inclusions
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Ground truth “arby1101”: fluorescent inclusion
[Rodriguez-Vega et al. 2026]

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Regularized least-squares reconstruction

Assuming Gaussian noise, $\varepsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$, the nonnegative maximum a posteriori (MAP) estimator reads:

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The first term is the **least-squares (LS)** data term; we compare three regularizers $R(\mathbf{c})$:

Regularizer	$R(\mathbf{c})$	Expected effect
ℓ_2	$\frac{1}{2}\|\mathbf{c}\|_2^2$	quadratic penalty: smooth solutions
ℓ_1	$\|\mathbf{c}\|_1$	sparsity penalty
TV	$\|\nabla \mathbf{c}\|_{2,1}$	total-variation penalty: piecewise-constant reconstructions

ℓ_2 and ℓ_1 regularization: variable-metric proximal gradient

For non-differentiable R, we rely on proximal splitting methods [Combettes & Pesquet 2011]. The variable-metric proximal gradient scheme [Park et al. 2019] iterates:

$$\mathbf{c}^{k+1} = \operatorname{prox}_{\mathbf{D}_k^{-1},\, \lambda R + \iota_{\mathbb{R}_+}} \left(\mathbf{c}^k - \mathbf{D}_k^{-1} \mathbf{A}^\top (\mathbf{A}\mathbf{c}^k - \mathbf{m}) \right)$$

$\mathbf{D}_k$: diagonal preconditioning matrix, estimated with a diagonal Barzilai–Borwein stepsize [Park et al. 2019], together with a nonmonotone line search [Grippo et al. 1986].

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LS + ℓ_2: adaptive diagonal preconditioning

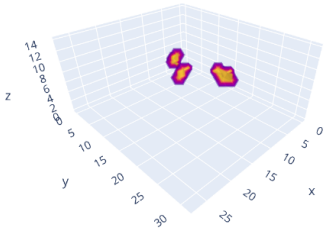
$$\mathbf{c}^{k+1} = P_+\Big[\mathbf{c}^k - \mathbf{D}_k^{-1}\big(\mathbf{A}^\top(\mathbf{A}\mathbf{c}^k - \mathbf{m}) + \lambda_2 \mathbf{c}^k\big)\Big]$$

LS + ℓ_1: adaptive diagonal proximal gradient

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P_+: projection onto the nonnegative orthant ($\mathbf{c} \geq 0$). For ℓ_1 and $\mathbf{c} \geq 0$, soft-thresholding reduces to a shift by λ_1 followed by P_+.

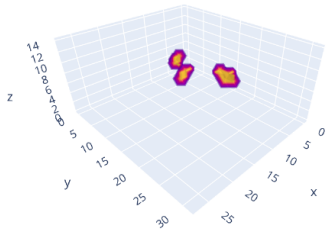
Results: LS + ℓ_2 and LS + ℓ_1



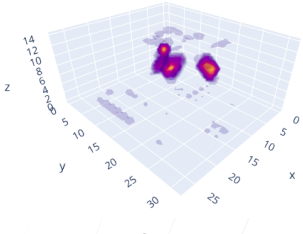
Reference: ground truth “arby1101”

Same LS data term for both; only the regularizer and the corresponding optimization update change.

Results: LS + ℓ_2 and LS + ℓ_1



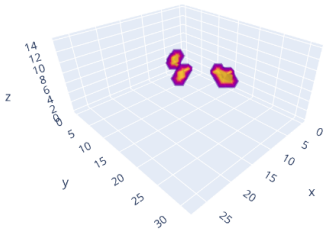
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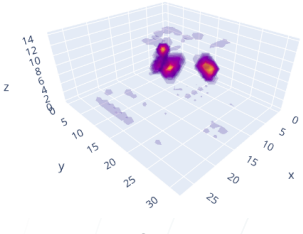
LS + ℓ_2
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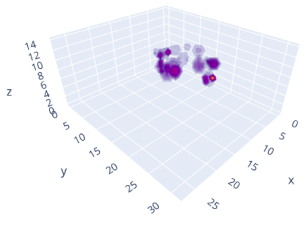
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LS + ℓ_2
pSNR = 29.4 dB



LS + ℓ_1
pSNR = 26.9 dB

Same LS data term for both; only the regularizer and the corresponding optimization update change.

Total-variation (TV) regularization

We replace $R(\mathbf{c})$ by the isotropic 3D total variation, which favors *piecewise-constant* reconstructions, well suited to fluorescent inclusions:

$$R(\mathbf{c}) = \|\nabla \mathbf{c}\|_{2,1} \;=\; \sum_i \|\nabla_i \mathbf{c}\|_2$$

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The TV term is non-smooth and coupled through ∇: we use a **primal–dual** scheme [Condat 2013, Chambolle & Pock 2011], with a data (gradient) step followed by a dual projection onto the ball $\|\mathbf{q}_i\|_2 \le \lambda_{\mathrm{TV}}$:

Primal–dual updates (data step + TV dual projection)

$$\mathbf{c}^{k+1} = P_+\Big[\mathbf{c}^k - \tau\big(\mathbf{A}^\top(\mathbf{A}\mathbf{c}^k - \mathbf{m}) + \nabla^\top \mathbf{q}^k\big)\Big]$$

$$\mathbf{q}^{k+1} = \Pi_{\|\mathbf{q}_i\|_2 \le \lambda_{\mathrm{TV}}}\Big(\mathbf{q}^k + \sigma \nabla \bar{\mathbf{c}}^{k+1}\Big), \qquad \bar{\mathbf{c}}^{k+1} = 2\mathbf{c}^{k+1} - \mathbf{c}^k$$

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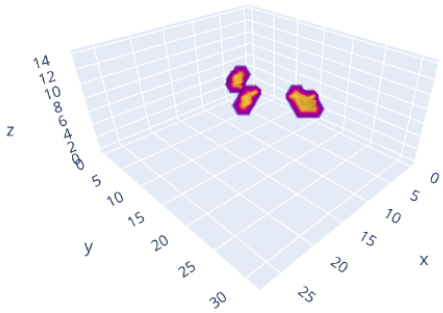
$$\mathbf{c}^{k+1} = P_+\Big[\mathbf{c}^k - \tau\big(\mathbf{A}^\top(\mathbf{A}\mathbf{c}^k - \mathbf{m}) + \nabla^\top \mathbf{q}^k\big)\Big]$$

$$\mathbf{q}^{k+1} = \Pi_{\|\mathbf{q}_i\|_2 \le \lambda_{\mathrm{TV}}}\Big(\mathbf{q}^k + \sigma \nabla \bar{\mathbf{c}}^{k+1}\Big), \qquad \bar{\mathbf{c}}^{k+1} = 2\mathbf{c}^{k+1} - \mathbf{c}^k$$

$\mathbf{q}$: dual variable (one 3D vector $\mathbf{q}_i$ per voxel); $\tau, \sigma > 0$: stepsizes; $\Pi_{\|\mathbf{q}_i\|_2 \le \lambda_{\mathrm{TV}}}$: projection of each $\mathbf{q}_i$ onto the ℓ_2-ball of radius λ_{TV}, i.e. $\mathbf{q}_i \mapsto \mathbf{q}_i / \max\big(1, \|\mathbf{q}_i\|_2 / \lambda_{\mathrm{TV}}\big)$.

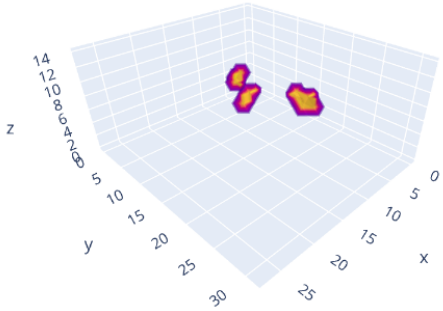
Convergence is guaranteed if $\frac{1}{\tau} - \sigma\|\nabla\|^2 \ge \frac{1}{2}\|\mathbf{A}\|^2$ [Condat 2013].

Results: LS + TV

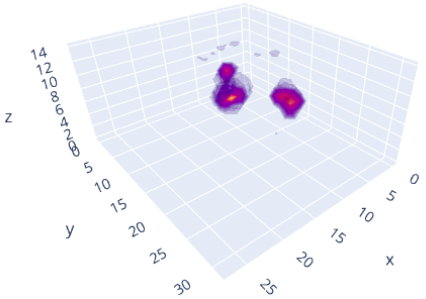


Reference: ground truth “arby1101”

Results: LS + TV

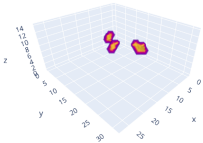


Reference: ground truth “arby1101”

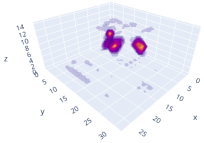


LS + TV pSNR = 29.3 dB

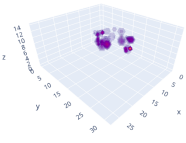
Comparison of the three regularizers



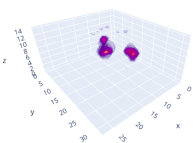
Reference



LS + ℓ_2

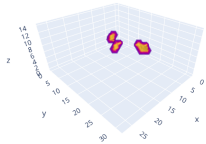


LS + ℓ_1

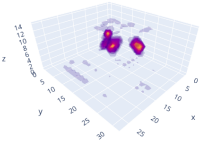


LS + TV

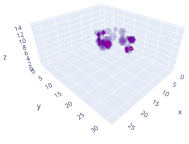
Comparison of the three regularizers



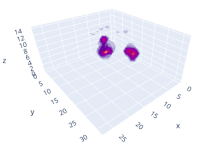
Reference



LS + ℓ_2

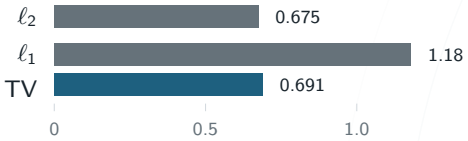


LS + ℓ_1

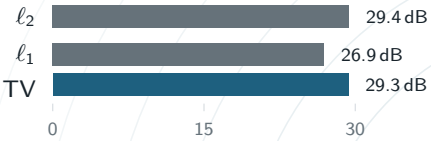


LS + TV

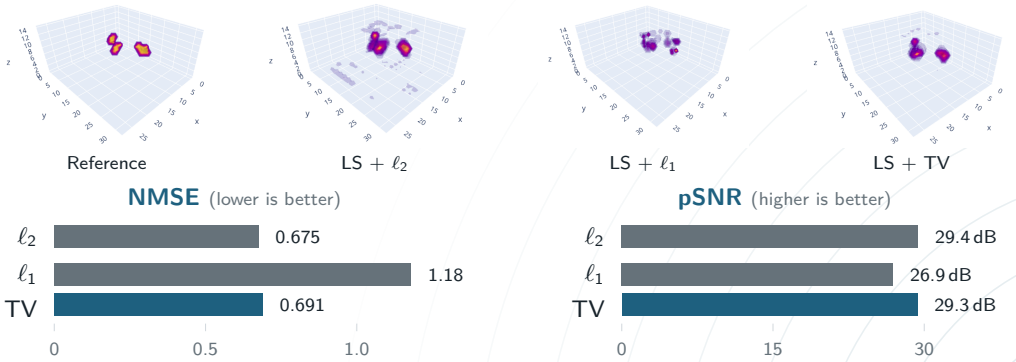
NMSE (lower is better)



pSNR (higher is better)



Comparison of the three regularizers



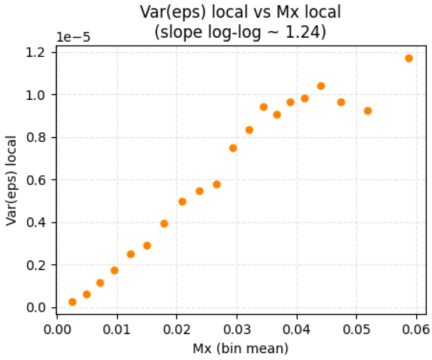
Same LS data term for the three reconstructions [Combettes & Pesquet 2011, Condat 2013]. Note that the best scores do not necessarily correspond to the best-looking reconstruction.

How can we stably invert a severely ill-conditioned fDOT operator?



Mean–variance diagnostic: signal-dependent variance

A power-law noise model writes $\mathrm{Var}(M) \propto [\mathbb{E}(M)]^p$. The exponent p describes how quickly variance grows with the mean: $p = 0$ is constant variance and $p = 1$ is Poisson-like scaling.



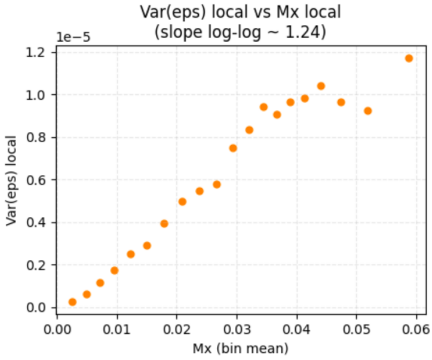
Empirical mean–variance diagnostic.

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A log–log fit estimates p from the data:

$$p \approx 1.24$$



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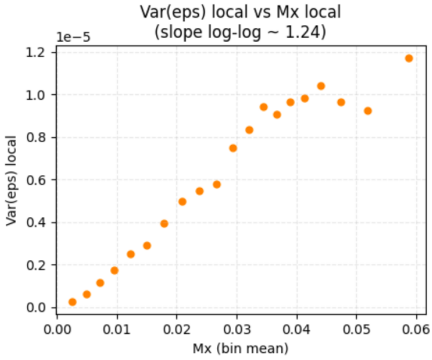
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This indicates **signal-dependent** (heteroscedastic) noise: the variance grows somewhat faster than in the Poisson-like case $p = 1$, and unlike constant-variance Gaussian noise ($p = 0$).



Empirical mean–variance diagnostic.

Poisson noise $\longrightarrow$ Kullback–Leibler (KL) fidelity

For independent Poisson measurements, $M_n \sim \text{Poisson}(\mu_n)$ and $\text{Var}(M_n) = \mu_n$. The negative log-likelihood yields a **KL** data-fidelity term. With TV regularization, we reconstruct $\mathbf{c}$ by minimizing [Chambolle & Pock 2011]:

KL data-fidelity term

$$D_{\mathrm{KL}}(\mathbf{m} \| \mathbf{Ac}) = \sum_m \Big[(\mathbf{Ac})_m - m_m \log \big((\mathbf{Ac})_m \big) \Big]$$

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KL + TV

$$\widehat{\mathbf{c}} = \underset{\mathbf{c}\geq 0}{\arg\min} \Big[D_{\mathrm{KL}}(\mathbf{m}\|\mathbf{Ac}) + \lambda_{\mathrm{TV}}\|\nabla\mathbf{c}\|_{2,1}\Big]$$

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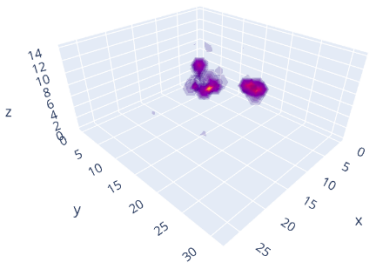
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KL + TV reconstruction
pSNR = 28.9 dB

Take-home messages and open question

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Take-home messages and open question

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Take-home messages and open question

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- **Our evaluation metrics are not satisfactory:** ℓ_2 scores best on NMSE and pSNR, yet TV gives visibly better reconstructions of the inclusions.
- The measured noise follows $\mathrm{Var}(M) \propto [\mathbb{E}(M)]^p$ with $p \approx 1.24$, outside the Gaussian ($p = 0$) and Poisson ($p = 1$) models.

Open question

The measured scaling $p \approx 1.24$ is not captured by the constant-variance Gaussian ($p = 0$) or Poisson ($p = 1$) models.

Can the data-fidelity term be learned from the observed error statistics?

R&D Imaging Seminar · 3 November 2026

Thank you for your attention!

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Aix-Marseille Université · Institut de Mathématiques de Marseille (I2M) · Institut Fresnel

Appendix — Generalization: Tweedie noise model

We relax the Gaussian assumption by modelling $\mathbf{m}$ with a **Tweedie distribution** [Févotte & Idier 2011], an exponential family parametrized by $p \in \mathbb{R}$ that includes Gaussian ($p = 0$), Poisson ($p = 1$) and Gamma ($p = 2$):

$$\mathbf{m} \sim \mathrm{Tw}_p(\mathbf{Ac},\, \sigma^2)$$

The MAP estimator is [Févotte & Idier 2011]:

$$\widehat{\mathbf{c}} \;\in\; \underset{\mathbf{c} \geq 0}{\arg\min}\; d_\beta(\mathbf{m} \,|\, \mathbf{Ac}) \;+\; \lambda\, R(\mathbf{c}), \qquad \beta = 2 - p$$

with the β-divergence:

$$d_\beta(\mathbf{m} \,|\, \hat{\mathbf{m}}) = \sum_i \frac{\hat{m}_i^{\,\beta}}{\beta(\beta-1)} - \frac{m_i\, \hat{m}_i^{\,\beta-1}}{\beta - 1} + \frac{m_i^{\,\beta}}{\beta}$$

β	Divergence	Noise
2	Least squares	Gaussian
1	Kullback–Leibler	Poisson
0	Itakura–Saito	Gamma

The measured $p \approx 1.24$ would lie between the Poisson ($\beta = 1$) and Gamma ($\beta = 0$) cases, on the Poisson side, i.e. $\beta \approx 0.76$ if this model is adopted.

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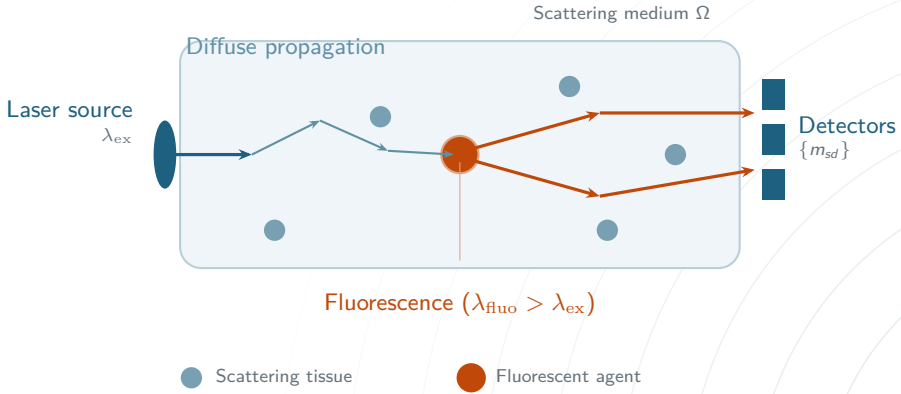
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C. Févotte and J. Idier, "Algorithms for nonneg. matrix factorization with the beta-divergence," *Neural Computation*, vol. 23, no. 9, pp. 2421–2456, 2011.

Appendix — Physical context (diagram)



Goal: reconstruct the fluorophore distribution inside Ω from surface measurements.

Appendix — Boundary conditions & Born approximation

Robin boundary conditions

On the boundary $\partial\Omega$, partial reflection of light (refractive-index mismatch) is accounted for:

$$\Phi + 2A_r\, D\,(\hat{\mathbf{n}} \cdot \nabla\Phi) = 0, \quad \mathbf{r} \in \partial\Omega$$

$\hat{\mathbf{n}}$: outward normal, A_r: reflection coefficient (refractive index of the medium).

Born approximation

We decompose $\Phi = \Phi^{(0)} + \Phi^{(s)}$ (homogeneous background + perturbation due to the fluorophore). To first order:

$$\Phi^{(s)}(\mathbf{r}_d) \approx \int_\Omega G(\mathbf{r}_d, \mathbf{r})\, \eta\, \mu_{af}(\mathbf{r})\, \Phi^{(0)}(\mathbf{r})\, \mathrm{d}\mathbf{r}$$

G: Green's function of the background medium,
valid if $\|\Phi^{(s)}\| \ll \|\Phi^{(0)}\|$ (low fluorophore concentration).

This linearization yields the matrix relation between measurements and fluorophore, $\mathbf{m} = \mathbf{A}\mathbf{c} + \varepsilon$, where $\mathbf{A}$ gathers the Green's functions weighted by $\Phi^{(0)}$ for each source–detector pair.

Appendix — β-NMF algorithm: multiplicative updates

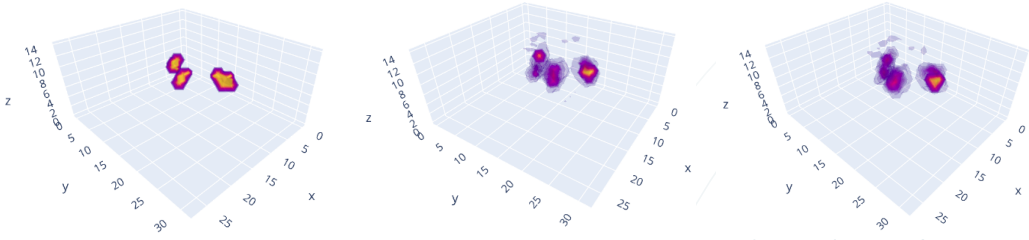
Majorization–minimization update [Févotte & Idier 2011]:

$$\mathbf{c}^{k+1} \;=\; \mathbf{c}^{k} \;\odot\; \left(\frac{\mathbf{A}^\top \left(\mathbf{m} \odot (\mathbf{A}\mathbf{c}^k)^{\odot(\beta-2)} \right)}{\mathbf{A}^\top (\mathbf{A}\mathbf{c}^k)^{\odot(\beta-1)} + \lambda_{\ell_1} + 2\lambda_{\ell_2}\, \mathbf{c}^k} \right)^{\gamma}$$

$\gamma = 1$ if $\beta \in [1, 2]$, $\quad \gamma = (2-\beta)^{-1}$ if $\beta < 1$, $\quad \gamma = (\beta-1)^{-1}$ if $\beta > 2$.

ℓ_1 (penalty λ_{ℓ_1}) and ℓ_2 (penalty λ_{ℓ_2}) regularizations are included in the denominator. Nonnegativity is preserved at every iteration.

Appendix — β-NMF results (arby1101)



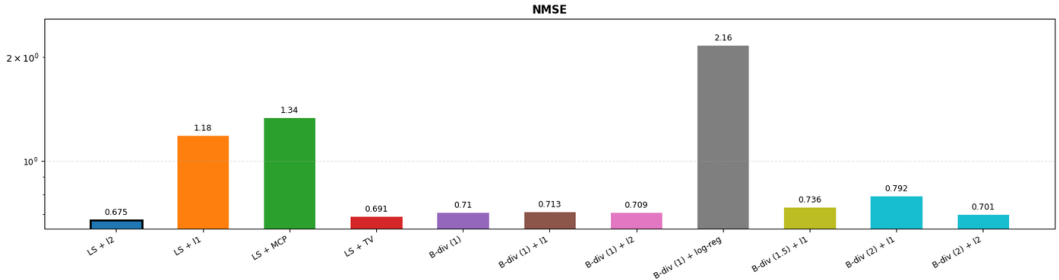
Ground truth

$\beta = 1$ (KL) + ℓ_1
pSNR = 29.1 dB

$\beta = 2$ (LS) + ℓ_1
pSNR = 28.7 dB

Influence of the parameter β on the reconstruction, simulated fDOT data.

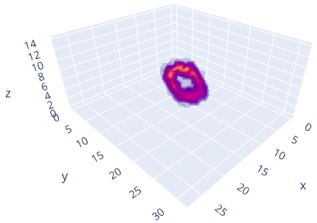
Appendix — NMSE of all methods (arby1101)



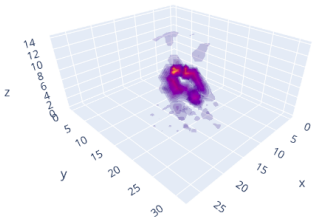
NMSE values for each method.

Compared models: FB splitting $\ell_2 + \ell_1/\ell_2$/MCP · Primal–dual LS + TV · MM β-divergence ($\beta = 1$) + ℓ_1/ℓ_2/Log. Data: arby1101.

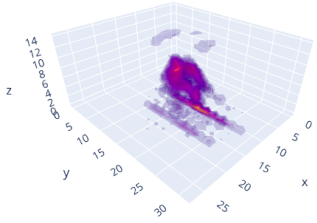
Appendix — Results on arby0313 (1/2)



Ground truth



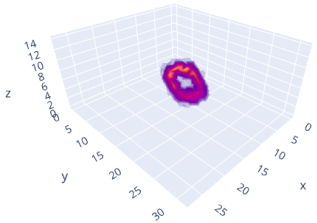
LS + TV
pSNR = 30.6 dB



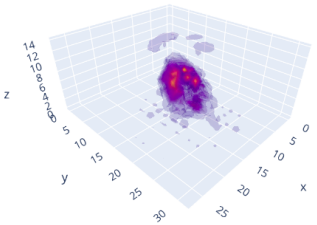
LS + ℓ_2
pSNR = 30.2 dB

Simulated fDOT data.

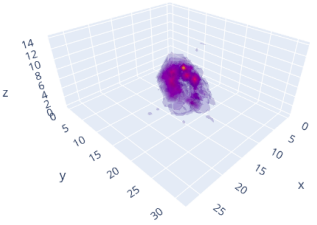
Appendix — Results on arby0313 (2/2)



Ground truth



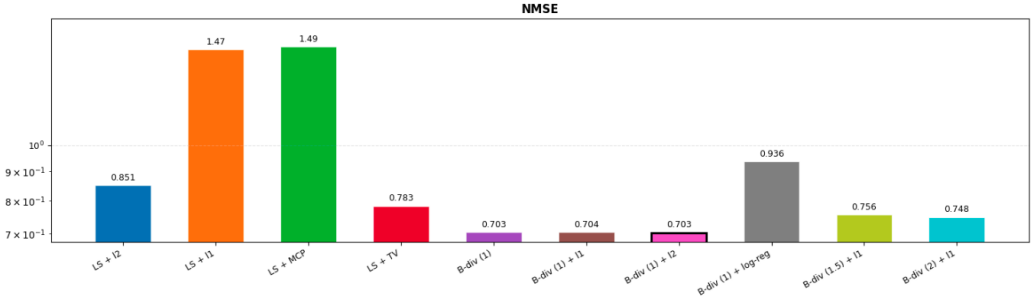
LS + ℓ_1
pSNR = 30.8 dB



KL + ℓ_2
pSNR = 31 dB

Simulated fDOT data.

Appendix — NMSE of all methods (arby0313)



NMSE values for each method.

Compared models: FB splitting $\ell_2 + \ell_1/\ell_2$/MCP · Primal–dual LS + TV · MM β-divergence ($\beta = 1$) + ℓ_1/ℓ_2/Log. Data: arby0313.